Cell Biology Homework Assignment 2

Name _____ Student ID_____

Part 1. Multiple choice.

More than one answer per question (2.5 point per each, total 25 points).

1. In which organelle(s) are proteins translated?
   A. Mitochondria
   B. Nucleus
   C. Golgi
   D. Endoplasmic Reticulum

2. Which of the following are part of the definition of the Cell Theory?
   A. All forms of life contain one or more cells.
   B. All functions that make life possible occur within cells.
   C. Viruses are the smallest unit of life.
   D. All cells come from pre-existing cells.
   E. Cells possess the genetic material required to regulate all cell functions.

3. Enzymes perform which of these functions: (_____)

   A. They increase the activation energy between reactants and products.
   B. They increase up the rates of a chemical reactions in cells
   C. They overcome larger activation energy thresholds.
   D. That they enable energetically unfavorable reactions.
   E. That they lower the activation energy between reactants and generation of products.

4. The driving forces for ion movement across a phospholipid membrane are due to:

   A. Chemical gradient
   B. Catalysis
   C. Electrical (charge) gradient
   D. Active transport
   E. Osmosis

5. The purpose(s) of the electron transport chain in mitochondria are

   A. To convert photon energy into electron transport.
   B. To generate a hydrogen gradient across the inner membrane.
   C. To convert CO_2 into a carbohydrate macromolecule.

D. To generate O_2
E. To convert chemical energy into the proton-motive force

6. The ribosome (_____)

A. Consists of 2 subunits

B. Is responsible for translating mRNAs

C. Is responsible for generating macromolecules of amino acids

D. Consists of RNA and proteins

E. Generates tRNAs

7. What is/are the following(s) NOT the function of the Golgi apparatus?

A. Transportation of protein molecules
B. Collection of protein molecules
C. Distribution of protein molecules
D. Digestion of protein molecules
E. Degradation of protein molecules

8. Which of the following factor(s) are the second messengers that are used for signal transduction? (_____)

F. cAMP
G. cGMP
H. Sodium ion
I. Calcium ion
J. IP_3
K. DAG
L. Small G-proteins

9. Which of the following are true for Arp2/3?

A. Involved in the formation of tight parallel bundles.

B. Establishing new branches of microfilaments

C. Binds to established microfilaments

D. Involved in the formation of dendritic network of microfilaments

E. Mimic actin monomers

10. Which are functions of the Endoplasmic Reticulum?

A. Give the cell mechanical strength.
B. Anchor/Support the organelles.

C. Glycosylate proteins.

D. Translate proteins that will be embedded into phospholipid membranes.

E. Translate proteins that will undergo exocytosis.

Part 2. Essay Questions (75 points).

Question 1. In muscle cells, a rise in intracellular Ca^{2+} triggers the contraction of the myosin–actin system. Meanwhile, muscle cells have specialized Ca^{2+} channels—called ryanodine receptors due to their sensitivity to the drug ryanodine—which are located in the membrane of the sarcoplasmic reticulum. In contrast to the IP_3-gated Ca^{2+} channels, Ca^{2+} itself can also open ryanodine receptors. Describe the roles and consequences of ryanodine channels in muscle cell contraction. (25 points)

Ca^{2+}-activated Ca^{2+} channels create a positive feedback loop: the more Ca^{2+} that is released, the more Ca^{2+} channels open (12.5 point).

The Ca^{2+} signal in the cytosol is therefore propagated explosively throughout the entire muscle cell, thereby ensuring that all myosin–actin filaments contract almost synchronously (12.5 point).

Question 2. Accurately diagram and correctly label the complete process that results in the generation of cAMP by a ligand for G-protein coupled receptors. (25 points)

Ligand binds to G-protein coupled receptor (1 point) on the extracellular side of the plasma membrane (3 points).

GPCR is a 7-pass transmembrane protein in the plasma membrane (3 point)

Ligand-bound GPCR interacts with the heterotrimeric G-protein complex that consist of Galpha-s (3 point for s designation), G-beta and G-gamma on the intracellular side of the plasma membrane. (7 points)

GPCR acts as a Guanine exchange factor to replace GDP with GTP on the Ga subunit. (3 point)

GTP-bound G-alpha separates from G-beta/gamma. (3 point)

GTP-bound G-alpha interacts with adenylyl cyclase. (3 point)

Active adenylyl cyclase converts ATP to cAMP. (3 point)

Question 3. Accurately diagram and correctly label the processing and activation of Notch signaling pathway. (25 points)

The Notch receptor undergoes three successive proteolytic cleavage steps, but only the last two depend on Delta binding (3 point).

As part of its normal biosynthesis, it is cleaved in the Golgi apparatus to form a heterodimer, which is then transported to the cell surface as the mature receptor (3 point).

Notch and Delta interact through their repeated EGF-like domains (3 point).

The binding of Delta to Notch induces a second cleavage in the extracellular domain (2 point), mediated by an extracellular protease (3 point).

A final cleavage quickly follows, cutting free the cytoplasmic tail of the activated receptor (3 point).

This final cleavage of the Notch tail occurs just within the transmembrane segment (2 point), and it is mediated by a protease complex called γ-secretase (3 point).

The released Notch tail migrates into the nucleus, where it binds to the Rbpsuh protein, which it converts from a transcriptional repressor to a transcriptional activator (3 point).